



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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Computer Fundamental

COMPUTER SOFTWARE

Q. What is Software ?

- Software is a set of program that control the working of the computer hardware.
- Program is a set of instruction.
- Software is the detailed instructions that control the operation of a computer system.
- Software is untouchable or intangible part of the computer.
- Smallest unit of software (S/W) – Module / Thread
- Person who develop software called – Programmer
- e.g – Operating System, Compiler, Ms word, Ms excel, MS Power point, Tally, Photoshop, Foxpro, MySQL, VLC media player, CorelDraw, Chrome, Mozilla Firefox, Safari....etc

❖ CHARACTERISTIC OF GOOD SOFTWARE

- Good software should be usable even by the inexperienced users.
- Good software should be reliable and dependable (विश्वसनीय एवं भरोसेमंद).
- Good software should function efficiently.
- Good software should have a "life-time" at least measured in years.
- Good software should provide an appropriate/good user interface(GUI).
- Good software should be "easily" maintained.
- Good software should be fastly and Quickly.

❖ FUNCTIONS OF SOFTWARE

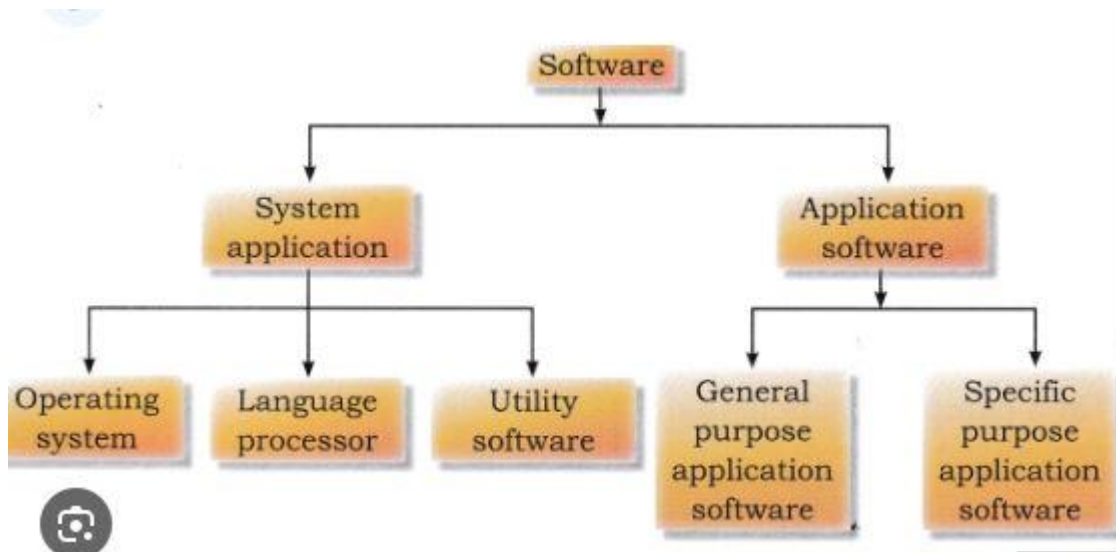
- Manage the computer resources of the organizations.
- Provide tools for people to take advantage of these resources.
- Easily communicate to hardware devices & storage devices.

❖ CLASSIFICATION OF COMPUTER SOFTWARE

Computer software can be classified into two categories-

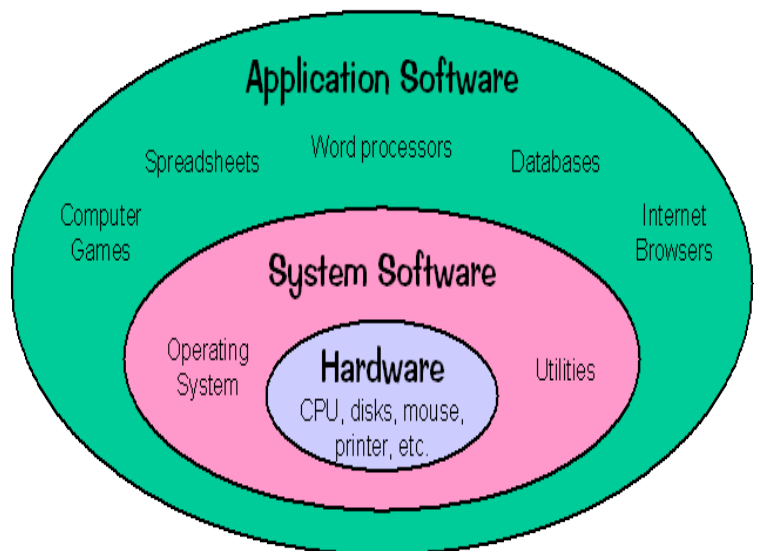
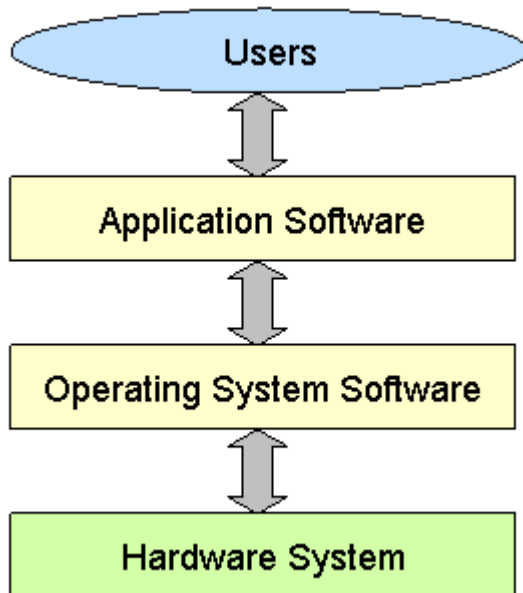
A. Systems Software

B. Application software



A. System Software / Application

System software are programs that control the any type of operations/Work in a computer and its devices. System software also serves as the interface between the user, the application software and hardware.



There are Four categories of systems software.

1. Operating system.
2. Language Translators.
3. Utility programs.
4. Device Drivers

1. Operating System

- An operating system (OS) is a set of programs containing instructions that co-ordinate all the activities among computer hardware devices.
- The operating system so much just like on the device driver to communicate with each hardware device.
- device driver is a small program that tells the OS how to communicate with the device.
- e.g –
 - ✓ MS-DOS - Owner Microsoft – Use PC
 - ✓ Android – Owner Google – Use smartphone / tablets
 - ✓ IOS – Owner Apple - Use smartphone / tablets
 - ✓ MACOS – Owner Apple – Use PC
 - ✓ Windows – Owner Microsoft – Use PC
 - ✓ Chrome OS – Owner google – Use tablets & netbooks.
 - ✓ Linux – Use PC - professionals, Cloud based services & websites.
 - ✓ Symbian – Owner Nokia – Use Smartphone
 - ✓ Ubuntu – Use PC



➤ Two types of OS based on User Interfaces

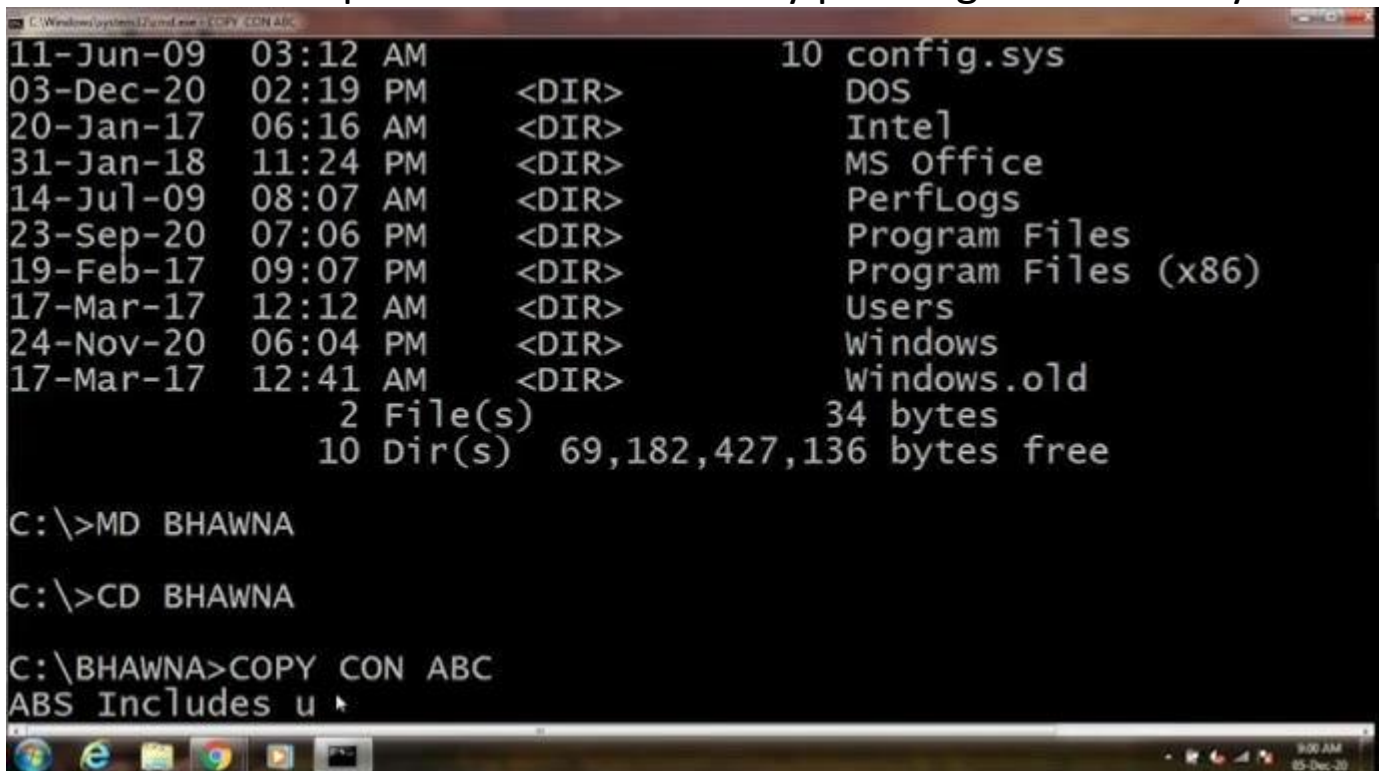
A. CLI (Command Line Interface)/ CUI (Command User Interface)

B. GUI (Graphical User Interface)

User Interfaces - This is the environment given by the OS to control how a user enters data and instructions and how information is displayed.

A. Command-Line Interface

- With a command line interface, a user types keyboard or press special keys on the keyboard to enter data and instructions.
- The set of commands a user uses to interact with the computer is called command language.
- Examples are DOS, UNIX, Linux.
- You are provided with a virtually empty screen with a blinking cursor where commands are keyed and the computer executes them by pressing the enter key.



```
C:\Windows\system32\cmd.exe - COPY CON ABC
11-Jun-09 03:12 AM          10 config.sys
03-Dec-20 02:19 PM          <DIR>      DOS
20-Jan-17 06:16 AM          <DIR>      Intel
31-Jan-18 11:24 PM          <DIR>      MS Office
14-Jul-09 08:07 AM          <DIR>      PerfLogs
23-Sep-20 07:06 PM          <DIR>      Program Files
19-Feb-17 09:07 PM          <DIR>      Program Files (x86)
17-Mar-17 12:12 AM          <DIR>      Users
24-Nov-20 06:04 PM          <DIR>      windows
17-Mar-17 12:41 AM          <DIR>      windows.old
                2 File(s)          34 bytes
            10 Dir(s) 69,182,427,136 bytes free

C:\>MD BHAWNA

C:\>CD BHAWNA

C:\BHAWNA>COPY CON ABC
ABS Includes u
```

Advantages of Command User Interface

- Commands are executed faster.
- Consumes limited Random Access Memory.

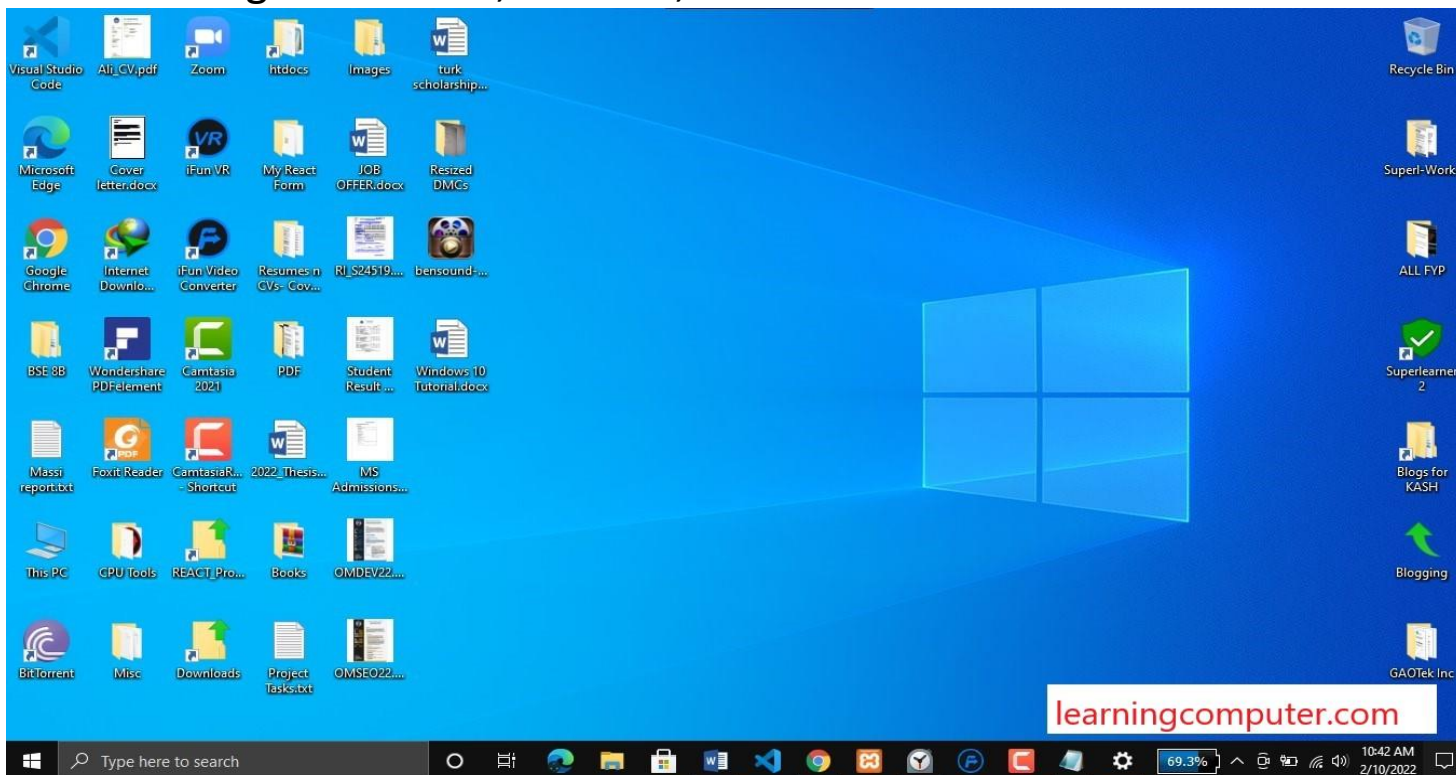
- Cannot easily affected by viruses.
- Ideal for programming and programmers.
- Commands can be grouped together.

Disadvantages of Command User Interfaces

- Commands are not easily memorized.
- Requires a lot of knowledge to master the commands.
Commands have to be typed correctly.
- No graphics on the screen.
- A mouse cannot be used

B. Graphical User Interface (GUI)

- Graphical user interface allows the user to use menus and visual images such as icons, buttons and other graphical objects to issue commands.
- E.g – Windows, Andriod,.....



Advantages of GUI

- It is user friendly because it is easy to learn and work with
- There is no need to type and memorize any command language.
- The interface is similar for any application.
- You are provided with a coloured screen with icons each representing a program.

- A mouse may be used.

Disadvantages of (GUI)

- Require much more memory and as well as the processor
- Require much more desk space to hold the files
- Difficult to automate for expert users
- Easily affected by viruses
- Not good for programmers.

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